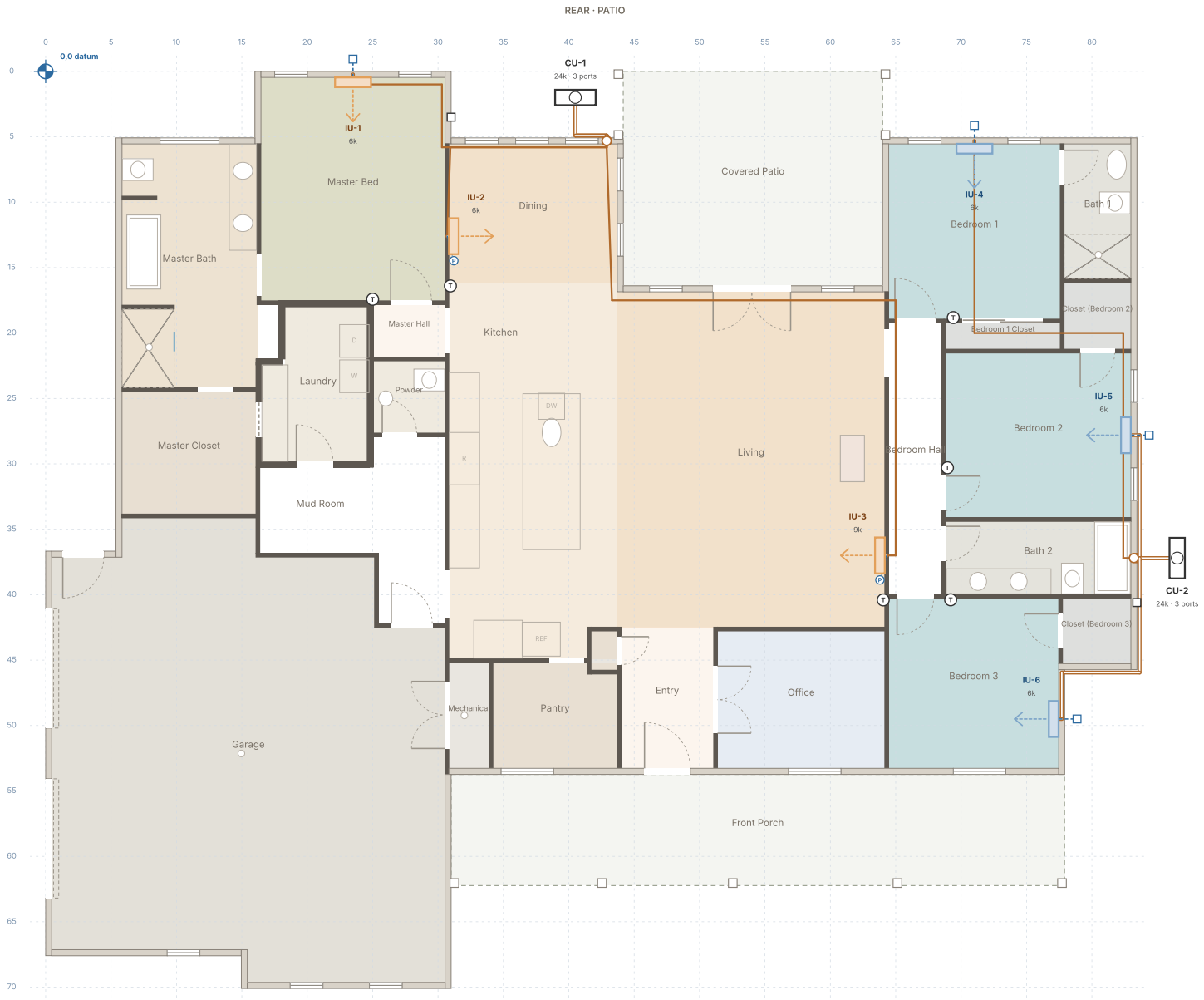


line set LS#, in the attic or inside a wall · line set outside, in a line-hide · sleeve through the wall · condensate · wall head on CU-1 IU# · wall head on CU-2 IU# · outdoor unit CU# · condensate pump · wired controller T#, 48" · disconnect DS#

Heads tinted by unit: CU-1 orange, CU-2 blue; shaded rooms are what each head serves. · CU-1 24k · CU-2 24k · cool 21 kBtu/h

draft

M-1 HVAC · 64 Stone Sheep Circle

Ductless: two cold-climate multi-zone mini-split heat pumps with wall heads; the radiant slab (R-1) carries the heat. App version v13. From the owner's plans (Larson House, 6/11/26).

Notes

1. Design conditions: Powell, WY (ASHRAE 2009, Cody Muni AWOS (nearest station to Powell), 99.6 % heating / 1 % cooling). Heating -11.4°F, cooling 87.8°F; inside 70°F / 75°F.
2. No ducts. The radiant slab (R-1) heats every room; the mini-splits cool the rooms with heads and give fast, second-source heat there. Rooms without a head (office, laundry, mud room, powder, halls) take air from the next room through their doors.
3. Sizing: each head carries the cooling load of the rooms it serves plus 15 % for latent and margin, with 6k the smallest head; the living head is held at 9k for the vault, the open kitchen and cooking. Each outdoor unit is the smallest common cold-climate multi-zone with enough ports, with the heads at no more than 130 % of its rating.
4. Outdoor units sit on a concrete pad against an exterior wall, raised 12" on a stand for snow, backs 30" off the wall so they are outside the eave drip line and can be serviced; snow guards on the roof above each. CU-1 is in the rear alcove between the master bedroom and the patio; CU-2 is on the Bath 2 wall of the bedroom wing. Line sets were routed for the shortest run from there.
5. Line sets to heads on exterior walls stay outside: from the unit in a painted line-hide along the wall at about 8' (just above the window heads) and straight through the wall into the back of the head (IU-5, IU-6). The others go through a sealed sleeve about 18" up, rise inside a full-height stud bay to the attic, run on the ceiling framing under the insulation and drop inside the wall behind the head. The living set rides the low edge of the kitchen/living vault at Y 17'-6", where the scissor trusses' bottom chord is only about 6" above the plate, then the Bedroom 1 and hall attics; nothing crosses the middle of the vault.
6. Heads mount high on the wall, top 6" below the ceiling, on a wall that lets them throw across the room. Exterior-wall heads drain condensate by gravity straight out the wall to a splash block (with the line set where there is one); interior-wall heads (IU-2, IU-3) get a mini condensate pump behind the head that discharges outside with the line set.

Open items

- CU-1 in the rear alcove: it is 2 ft from the edge of the covered patio and under the dining windows, so it is heard on the patio and in the dining room when it runs. The alternatives are the master bedroom's left wall (quiet, but the living set runs about 105 ft, past the per-head limit) or the front porch (every set under 75 ft, in sight of the street).
- CU-2 on the Bath 2 wall: 3 ft from Bedroom 2's window W14. The Bath 1 wall is the quiet spot (8 ft from any bedroom window) for about 10 ft more line.
- Exact models: pick the brand, then confirm the heads are an allowed combination and the line lengths are inside the unit's limits (CU-1: longest 69 ft, total 145 ft; CU-2: longest 57 ft, total 102 ft).
- Truss maker: the living set crosses the scissor trusses at Y 17'-6"; leave web openings for a 2" bundle there, or run it along a truss.
- Office: no head as drawn (the ask was the right-side bedrooms). A 6k head on the office's right wall would make CU-2 a 4-port.
- Bed walls: the master and Bedroom 1 heads sit on the rear wall between the windows, where a headboard may go. A head over a headboard works with its vanes set level; otherwise it moves to a side wall. Confirm furniture walls before rough-in.
- Ventilation: continuous bath fans (drawn) or paired through-wall ERVs (about 133 CFM continuous under the IRC).

7. Pressure test, evacuate and pull a micron vacuum per the maker. Every sleeve is sealed and flashed; line-hide painted to the siding.
8. Each outdoor unit has a dedicated 240 V circuit and a fused disconnect on the wall within sight (DS1 on the master bedroom's alcove wall, DS2 beside CU-2 on the Bath 2 wall), sized to the nameplate MCA/MOCP once the units are picked (E-1). The heads are powered from their outdoor unit through the 14/4 interconnect run with the line set.
9. Wired wall controllers T1-T6 at 48" by each door; set the heads to cool and to heat only below the slab's setpoint so they don't fight the radiant.
10. Ventilation: with no ducts, the draft assumes continuous low-speed bath fans (exhaust-only, E-1) for the IRC whole-house rate; paired through-wall ERVs are the upgrade (see open items).

Indoor heads

Tag	Room	Unit	Size	Position	Cool Btu/h	Line set	Condensate	Serves
IU-1	Master Bed	CU-1	6k	on the rear wall, centred 7'-1/2" from the left wall; top at 8'-6" (6" below the ceiling)	3,439	1/4" x 3/8" · 41 ft	gravity	Master Bed, Master Bath, Master Closet, Master Hall
IU-2	Dining	CU-1	6k	on the left wall, centred 7'-1/2" from the rear wall; top at 8'-6" (6" below the ceiling)	3,485	1/4" x 3/8" · 34 ft	pump	Dining
IU-3	Living	CU-1	9k	on the right wall, centred 20'-2" from the rear wall; top at 8'-6" (6" below the ceiling)	7,377	1/4" x 3/8" · 69 ft	pump	Living, Kitchen, Entry, Pantry
IU-4	Bedroom 1	CU-2	6k	on the rear wall, centred 6'-6 1/2" from the left wall; top at 8'-6" (6" below the ceiling)	2,263	1/4" x 3/8" · 57 ft	gravity	Bedroom 1, Bedroom 1 Closet, Bath 1
IU-5	Bedroom 2	CU-2	6k	on the right wall, centred 6'-3 1/2" from the rear wall; top at 8'-6" (6" below the ceiling)	2,149	1/4" x 3/8" · 19 ft	gravity	Bedroom 2, Closet (Bedroom 2), Bath 2
IU-6	Bedroom 3	CU-2	6k	on the right wall, centred 4'-3" from the closet wall; top at 8'-6" (6" below the ceiling)	1,797	1/4" x 3/8" · 27 ft	gravity	Bedroom 3, Closet (Bedroom 3)

Outdoor units

Tag	Location	Size	Heads connected	Cool block Btu/h	Heat at 5°F / design	Line sets
CU-1	Rear alcove, against the dining room's rear wall	24k, 3 ports	21k (88 %)	12,871	24,000 / 18,000	145 ft, longest 69
CU-2	Right side yard, on the Bath 2 wall	24k, 3 ports	18k (75 %)	5,588	24,000 / 18,000	102 ft, longest 57

Line sets

Set	Run	Size	Length	Outside	Route
LS-1	CU-1 → IU-1	1/4" x 3/8"	41 ft	6 ft	line-hide to the sleeve by the patio corner, up inside the wall, across the dining and master attics, down the rear wall
LS-2	CU-1 → IU-2	1/4" x 3/8"	34 ft	5 ft	line-hide to the sleeve by the patio corner, up inside the wall, across the dining attic, down the master/dining wall
LS-3	CU-1 → IU-3	1/4" x 3/8"	69 ft	6 ft	line-hide to the sleeve by the patio corner, up inside the wall, along the low edge of the vault into the Bedroom 1 and hall attics, down the living room's right wall
LS-4	CU-2 → IU-4	1/4" x 3/8"	57 ft	3 ft	sleeve behind the unit, up inside the wall, across the bedroom-wing attic, down the rear wall
LS-5	CU-2 → IU-5	1/4" x 3/8"	19 ft	19 ft	outside in a line-hide up and along the right wall, straight into the head
LS-6	CU-2 → IU-6	1/4" x 3/8"	27 ft	27 ft	outside in a line-hide along the right wall, around the closet's two corners, straight into the head

Room loads

Room	Floor sf	Heat Btu/h	Cool Btu/h	Head
Master Bed	240	2,785	1,725	IU-1
Master Bath	192	3,455	960	IU-1
Master Closet	96	1,363	271	IU-1
Master Hall	22	108	34	IU-1
Laundry	87	904	587	— (spill from next room)
Powder	30	150	47	— (spill from next room)
Mud Room	133	1,236	356	— (spill from next room)
Dining	136	3,771	3,031	IU-2
Kitchen	370	2,280	2,370	IU-3
Pantry	77	1,441	694	IU-3
Mechanical	25	493	99	— (spill from next room)
Living	524	5,927	3,095	IU-3
Entry	76	1,223	256	IU-3
Coat Closet	6	30	9	— (spill from next room)
Office	134	1,977	1,571	— (spill from next room)
Bedroom Hall	86	422	133	— (spill from next room)
Bedroom 1	176	3,424	1,729	IU-4
Bedroom 1 Closet	18	86	27	IU-4

Room	Floor sf	Heat Btu/h	Cool Btu/h	Head
Bath 1	53	1,443	211	IU-4
Closet (Bedroom 2)	26	521	84	IU-5
Bedroom 2	178	2,363	1,615	IU-5
Bath 2	79	817	170	IU-5
Bedroom 3	169	3,163	1,436	IU-6
Closet (Bedroom 3)	26	916	127	IU-6

System

Item	Size	Basis
Heating load	40,299 Btu/h	70°F inside, -11.4°F outside; carried by the radiant slab (R-1)
Cooling load	20,637 Btu/h sensible	75°F inside, 87.8°F outside
Outdoor units	CU-1 24k + CU-2 24k	cold-climate multi-zone, sized on cooling and the head count
Mini-split heat	36,000 Btu/h at -11°F	backup to the slab in the rooms with heads
Indoor heads	6	high-wall, 6" below the ceiling
Line sets	247 ft	longest 69 ft; 65 ft of it outside in line-hide

Bill of materials

Rough-order quantities from the layout, for pricing. The HVAC contractor takes off the final list.

Item	Qty	Notes
Outdoor unit, 24k 3-port	1	CU-1, cold-climate multi-zone
Outdoor unit, 24k 3-port	1	CU-2, cold-climate multi-zone
Wall head, 6k	5	
Wall head, 9k	1	
Line set, insulated	272 ft	all sets + 10 %; sizes per the line-set table
Interconnect cable 14/4	272 ft	with the line sets
Line-hide cover	72 ft	painted to the siding
Wall sleeves	6	sealed and flashed
Condensate pump	2	IU-2, IU-3
Wired wall controller	6	T1-T6
Pad, stand, snow guards	2	12" stands on pads
Disconnect	2	DS1, DS2; circuits on E-1